

Notes on decentralizing digital transformation projects and its possible impact on sustainability

A semi-automated genAI-snowballing literature review

Abstract— This article examines whether decentralization—through generally purposed blockchain, Distributed Ledger Technologies (DLTs), and Decentralized Autonomous Organizations (DAOs) support the long-term sustainability (in terms of stability and resilience) of digital transformation projects both in the means of development and management. Using a semi-automated snowballing literature review with general AI screening agent, two rounds of citation tracing were performed from six seed articles, gathering 1577 candidates. After filtering, 26 high-relevance papers were analyzed. A binomial test on the results suggests, at 10% error tolerance, that current literature is shifting toward recognizing decentralization as a key factor in sustainable digital transformation. Open research directions are identified for further investigation.

Keywords—digital transformation, decentralization, sustainability, blockchain, distributed ledger technology, systematic literature review, snowballing, generative AI

I. INTRODUCTION

A. Context

Digital transformation (DT) as process both in context of management and of development was reshaping organizations and institutions on a huge scale in many countries. There exists a significant group of technological tools involving generally purposed blockchains, distributed ledger technologies (DLTs), decentralized autonomous organizations (DAOs) which are nowadays setting trends in architecture choices and governance within that transformation. On the other hand, since those are relatively new, the demand on sustainability, in the sense of long-term stability and resilience both before and after releases of projects, is increasing. This group of three is converging, but rarely studied together in one context.

B. Motivation

After surfacely and fairly randomly perusing the existing information on the matter (sites, forums, developer chats, conferences notes, etc.) it became obvious that these concepts are treated in pairs at best – digital transformation and sustainability, or blockchain and governance. But rarely the the focus is on the triad itself. Additionally, there is no consolidated view of how the decentralization specifically influence the sustainability in previously explained manner. This gap the article addresses.

C. Objectives & Research Question

The research question is as follows: “Does decentralization in digital transformation impact sustainability as stable long-term development?”

And the objectives this article aims to reach:

- Perform utilizing a generative AI agent a snowballing research of existing literature.
- Synthesize what the existing literature says about the relationship inside the above mentioned triad.
- And identify where an independent researcher could continue his or her research to further deepen the understanding of the current narrative in the academic field.

II. EXISTING CONCEPTS

The following is a brief cover up of separate concepts of the triad, to show the components which are going to be relied upon heavily in the synthesis of results:

A. Decentralization

In terms of management and development processes this concept usually applied in the manner of organizing the project into a DAO utilizing trustless smart-contract based agreements of distribution and redistribution of access to utility functions both of a project and a product under development Transaction cost theory described in [30] and [31] suggests among other ideas that the mentioned means are quickly adopted not only because of the efficiency of the trustless technology, but as a consequence of trustlessness – ability not to synthesize additional processes validation mechanisms both in development and management of organizations. Institutional theory presented in [28] and [29] informs its reader that the decentralization tools are widely accepted by both governmental and industrial structures. And finally commons governance described in [32] put a lot of faith in the trustlessness which comes with current existing tools of decentralization.

B. Digital Transformation

This concept is fairly good covered by [33] for example. However it is worth mentioning that this process have changed not just tools of project development and management, but restructured how value is created, teams

competed and decisions made. Namely, [34] argues that because of the process business strategy is now mixed with development strategies.

C. Sustainability

Sustainability as stability and resilience in project and product development is commonly confused with sustainability in the terms of ecological concerns and green economy. But in this triad the concept is well illustrated by the fact that many big tech companies which have a direct influence on many countries covering up most GDP on the planet have developed a habit to utilize concept of stakeholders in collaborative governance as described in [36] to achieve sustainable enterprise performance [35] long-term.

III. METHODOLOGY

A. Framing the research question

In context of methods utilized the research question (RQ) is now sounding like: “Given certain subset of starting articles and performing systematic literature review, do we have a clear view, that new articles found are indicating a shift of narrative among researchers towards describing decentralization as a key factor in digital transformation to sustain product results and their support?”

In other words, no hypothesis’ tested. Only field mapping.

B. Applied research method

For this systematic literature review (SLR) a snowballing method [37] was performed. Main reason for this method in favor is that since the RQ is more about finding trends and leads for further research, the citations are considered more useful information-wise, rather than chosen terminology. As if it was in keywords based database search and classification.

Primary constraints for SLR (Inclusion/Exclusion logic):

1. The journals must be peer-reviewed or subjectively high importance (in authors views).
2. Language: only English.
3. Published after 2015, because at the eve and after DeFi summer (boom of Decentralized Finance projects) there have been produced a lot of subjectively meaningful material to research.
4. Sustainability interpreted as stability or resilience, rather than ecological feature (mining concerns, prices of video processors spikes, etc.).
5. Addressing at least two concepts of the triad.
6. The overall relevance analyzed by genAI is high.
7. Source is accessible online and distributed free.
8. Has DOI.

The role of genAI was as follows:

TABLE I. DIVISION OF LABOR WITH AUTONOMOUS AI AGENT

Order	Tasks and actors		
	<i>Task performed</i>	<i>Manually</i>	<i>By AI^a</i>
1	Retrieving "Cited by" lists from Google Scholar	Yes	No
2	Pasting candidate lists into chat with the agent	Yes	No
3	Screening candidates against criteria	No	Yes

Order	Tasks and actors		
	<i>Task performed</i>	<i>Manually</i>	<i>By AI^a</i>
4	Checking indexing and free access	No	Yes
5	Synthesizing arguments	Yes	Partly ^b

^a. The core for the agent was Claude Sonnet 4.6

^b. AI was helping with an interpretation of unknown terminology and unusually large conclusions

Additionally to Table I, another subjective rule was put down: the Rule of 7:

- For Round 2 the 7 most relevant of Round 1 Backward papers were used as seeds, with first 7 pages of Google Scholar citing results retrieved per paper.

The Rule was utilized for time optimization of manual tasks.

C. Data collection

The following is a selection procedure description:

1. 6 papers selected subjectively to cover the start of conceptual space:
 - a. An article about impacts of digital transformation in energy sector and their impacts on economy as a whole [1].
 - b. Utilizing DT in highly distributed (from perspective both resources and management) logistics and energy [2].
 - c. A discussion about impacts of DT on sustainability [3].
 - d. Implementation of DT with blockchain in food sector discussion [4].
 - e. A narrative about innovative intermediaries with communication over blockchain [5].
 - f. An E-Governance through AI and controlled by trustlessness of blockchain paper [6].
2. Round 1 of snowballing papers collected:
 - a. Forward: 302 articles.
 - b. Backward: 415 articles.
3. Round 2 of snowballing papers collected:
 - a. Forward: 513 articles.
 - b. Backward: 340 articles.
4. Total: 1577 articles.
5. Total included after screening: 128 articles.
 - a. From them: 75 of high relevance;
 - b. And 26 of them passed all the filtering criteria.
6. Saturation signal observed in Round 2 Backward: ~2% inclusion rate.

For systematizing collected citations and articles a tracker web application was developed [29].

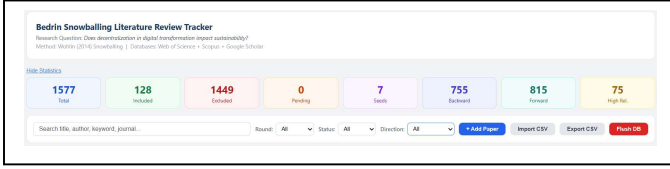


Fig. 1. Bedrin Snowballing Literature Review Tracker header screenshot

Fig. 1 shows a screenshot of header of the tracker illustrating the global statistics of data collected.

D. Analysis Method

As stated in Fig. 1, 128 papers were included, and then by applying two additional filters which were described in points B.6 and B.8 – confirmed DOI (ensuring full traceability and published status) and high subjective relevance semantics of an article which was assigned during the filtering process, this left 26 papers as the final analytical sample.

Synthesis based on the narrative had to be performed with one additional constraint: the conclusion section of each paper was interpreted as the primary unit of analysis. Conclusions usually contain papers final thoughts and rewinds of the narrative which is sufficient for identifying patterns and any other useful for the RQ information.

The analysis method is as follows: reading the conclusions of all final 26, identifying levels of relevance to the triad of DT, decentralization and sustainability, and trying to unify general narrative, or average narrative. While the levels of relevance are defined, rank them and systematize them into a table (Table II). After this systematizing treat gathered data as a tuple of random variables. Chosen ranking mechanism is classic as in any statistical analysis. Finally, explore them on the matter of belonging to the triad assuming their distribution is binomial. More about that further.

IV. OBTAINING AND ILLUSTRATING RESULTS

The ranking mechanism: if one concept of the triad taking place among other main concepts of a paper, then – 1, otherwise – 0. Then the resulting variation series are analyzed as a tuple of 3 binomial random variables, and the resulting sample is utilized to calculate probability that all 3 variables converging to 1.

To do that, the tuples are tested with help of exact one-sided binomial test to determine if general distribution truly converging to a tuple of ones with predefined error probability according to (1).

$$\frac{t}{n} > p_0 + z_{1-\alpha} * \sqrt{\frac{p_0^2}{n}} \quad (1)$$

Where: t – amount of tuples that are fully ones, n – sample size, p_0 – fraction of fully ones tuples in general binomial distribution, $z_{1-\alpha}$ – quantile of normal distribution, and α – is probability of error in the conclusion of the test.

TABLE II. TRIAD CONCEPTS REPRESENTED IN ANALYTICAL SAMPLE OF CONCLUSIONS

Paper conclusion resume	Triad		
	DT^a	D^b	S^c
Presented work-in-progress use case of using blockchain in egg supply chains [7].	1	1	1
Blockchain-based P2P energy trading making good promises but from policy makers point of view – dangerous [8].	1	1	1
Blockchain and Industry 4.0 technologies offer good potential, but should it be implemented incorrectly – severe privacy concerns play out [9].	1	1	1
States that ICOs partially democratize entrepreneurial finance by offering ethnic minorities better access to funding [10].	1	1	0
Concludes that blockchain technology significantly enhances green supply chain practices and organizational resilience management (ORM) in Pakistan and China's manufacturing sectors [11].	1	1	1
States that blockchain technology revolutionizing supply chain sustainability across economic, environmental, and social dimensions, but remains in early stages [12].	1	1	1
Concludes that despite being largely absent from e-Government literature, Bitcoin and blockchain technology show significant potential for public sector innovation [13].	1	1	1
Presents significant sociotechnical opportunities of blockchain across multiple industries, but their successful adoption in Europe depends on navigating complex policies [14].	1	1	1
Basically states that blockchain technology offers significant opportunities for businesses and policy makers [15].	1	1	1
Concludes that Middle East is advancing in its DT using decentralization through blockchain ecosystems [16].	1	1	0
Explains how blockchain technology could effectively address India's land record management challenges in a face of natural disasters [17].	1	1	1
Shows how Blockchain 3.0, combined with AI, can significantly enhance E-Government 3.0 applications by providing secure, decentralized, and transparent solutions for public services [18].	1	1	0
Illustrates that blockchain-based cryptosecurities offer a transparent, peer-to-peer alternative trading system that eliminates intermediaries	0	1	0

Paper conclusion resume	Triad		
	DT ^a	D ^b	S ^c
and reduces secrecy in financial markets [19].			
Tells that decentralizing technology adoption in Jordanian banks significantly improves both the quality of accounting information and the effectiveness of corporate governance [20].	1	1	1
Demonstrates that knowledge management significantly drives blockchain adoption and supply chain performance in Chinese manufacturing industries [21].	1	1	1
States that decentralization needs further research with quantitative metrics is needed to definitively assess whether its benefits outweigh its drawbacks [22].	1	1	1
Presents AI, blockchain, cloud computing, and data analytics (ABCD) as interconnected foundational technologies driving digital business transformation [23].	1	0	1
Shows significant promise for improving transparency and trust in decentralised rural governance, but requiring extensive research, before it can become a viable governance solution [24].	1	1	1
States that firms actively pursuing digital transformation through blockchain technology experience lower principal-agent conflicts and more engaged shareholders [25].	1	1	0
Shows that it is just another SLR on DT and Sustainability, but for policy maker specialists [26].	1	0	1

^a. Digital Transformation

^b. Decentralization

^c. Sustainability

V. TEST RESULT

First, let us fix some of the constants of the exact one-sided binomial tests: $t = 13$ (amount of tuples $\{1, 1, 1\}$), $n = 20$, $p_0 = 0.5$, $z_{1-\alpha} = 1.2816$, and $\alpha = 0.1$.

The final test result is: $0.65 > 0.6432$.

Which means that with chance of 10% of being wrong the above-described sample represents all other possible articles narratives that could be picked up utilizing the methodology that was also described above.

VI. CONCLUSION

This article set out to answer one question: does decentralization help digital transformation projects stay stable and successful long-term both in the means of development and management? To find out, a snowballing literature review was performed with general AI assistance, starting from six articles and screening over 1500 papers, ending up with 26 that were relevant enough to analyze.

After reading and coding the conclusions of those 26 papers, an exact binomial one-sided test was applied. The result shows that with a reasonable confidence level of 10%, the reviewed literature does lean toward saying yes – decentralization is increasingly seen by researchers as something that positively contributes to sustainable digital transformation. Meaning, if a researcher would want to repeat the described steps of snowballing and then filtering, with confidence level of 90% he or she would see the same trend as it was described here.

Blockchain technology as mean of decentralization keeps appearing across very different fields – supply chains, banking, governance, agriculture, energy – always in the role of making systems more transparent, resilient, and trustless. This consistency across sectors makes the narrative stronger.

The literature, however, is not yet conclusive. Several papers explicitly call for more quantitative research before any strong claims can be made. This review draws a map of the field rather than proves a theory.

For future researchers, the most obvious gaps are: measuring decentralization's impact numerically, studying DAO-governed projects over longer timeframes, and exploring how policy and policy making and their implementations shape adoption of decentralization.

All in all, the trend is real, the tools are there, but the academic field still has significant ground to cover.

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